Diego Macías Vega

Final-year science student in Madrid · physics, AI and engineering

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PROFILE

I build software on physics and mathematics and check it against measurement: reproduced baselines, validation suites and published negative results. I work with AI coding assistants and say so: the code is largely AI-written; the questions, the rules each project obeys, the decisions and the responsibility for every claim are mine.

EDUCATION

Bachillerato in Sciences and Technology IES San Agustín del Guadalix, Madrid — 2025 – 2027

- Mathematics II, Physics, Technology and Engineering, Computer Science. Final year; graduating June 2027.

University applications, September 2027 entry the Netherlands — in preparation

- Computer Science and Engineering (TU Delft, selection test) · Applied Physics (TU Eindhoven) · Advanced Technology (University of Twente).

CS50x: Introduction to Computer Science Harvard University, online — Started 2026

PROJECTS

RAAN · Open Orbital Intelligence Lab Python, TypeScript, WebGL, LightGBM — 2026

- Reproducible lab on public orbital data. Digital twin of 20,152 catalogued objects propagated with SGP4; propagator checked against Vallado's 666 reference vectors (max difference 3.5e-5 km).
- ESA Collision Avoidance Challenge: naive forecast reproduced exactly (L = 0.694). Best model scores 0.627 on the test set but fails cross-validation; reported as a negative result.
- Manoeuvre detection on 15 satellites, leave-one-satellite-out: F1 0.81 (classifier), 0.74 (physics baseline), 0.09 (chance). 6-page technical report.
- Code mostly written by an AI coding assistant and reviewed by an independent session; I set the question and the rules and answer for every claim.

Syndrome Lab quantum error correction — 2026, in progress

- Neural decoders vs minimum-weight perfect matching on the surface code, built with Stim and PyMatching. Results and report to be published late November 2026.

gravity-lab C++17, numerical methods — 2026

- N-body simulator of the Solar System validated against measurement: reproduces Mercury's relativistic perihelion precession (42.81 arcsec/century; accepted 42.98, error 0.4 %).
- Four integrators compared; Yoshida 4 (symplectic) holds energy error at 1.4e-11 over 100 years. 31 validation cases in CI. Browser build online.

KAIROS Python, FastAPI, PostgreSQL + pgvector, Docker — 2026

- Local-first personal AI assistant: 17 agents, 300+ tests, sandboxed self-modification that never merges without my approval. Architecture and decisions mine (77 written decision records); Claude as pair programmer.

Other — 2026

- Chess trainer (Stockfish and Maia, explanations fact-checked by test) · Quant Path (drills and 110 problems for quantitative trading, 101 verified by code) · TraceGuard (firmware changes checked against safety rules; paused MVP) · Sympto+ (multilingual health-information web app; paused).

EXPERIENCE

Private tutor Madrid — ongoing

- One-to-one tuition for school students.

SKILLS, LANGUAGES AND OTHER

Technical: Python, C++17, TypeScript; numerical integration, orbital propagation (SGP4), gradient boosting and recurrent networks, evaluation design (baselines, cross-validation, bootstrap); Git, Linux/WSL, Docker, GitHub Actions.

Languages: Spanish (native) · English, Cambridge C1 Advanced (score 186, obtained at 16) · Chinese, YCT 2.

Interests: superintelligence and robotics · physics, quantum above all · machine learning and AI · mathematics.

Sport: Judo, black belt.